

## **Hematology-Oncology Morbidity & Mortality Conference**

**DATE:** March 5, 2023

**PRESENTER:** Dr. B. Carter

**CASE:** Alan Watson (\*19.12.49, MRN AML075), a 71-year-old male with secondary-type AML refractory to hypomethylating agent therapy.

### **HISTORY OF PRESENT ILLNESS:**

Mr. Watson was a 71-year-old retired accountant with a history of hypertension who was referred to our clinic in July 2021 for evaluation of chronic macrocytic anemia and thrombocytopenia that had been slowly progressing over two years. This antecedent history was highly suspicious for an underlying myelodysplastic syndrome (MDS). A bone marrow biopsy on August 3, 2021, confirmed the diagnosis had transformed to Acute Myeloid Leukemia, with 35% myeloblasts. The marrow showed prominent trilineage dysplasia.

### **Key Diagnostic Features:**

- **Molecular Profile:** Next-generation sequencing was critical. It revealed mutations in **BCOR** and **U2AF1**. These are spliceosome and chromatin regulator mutations commonly seen in MDS and are recognized markers of AML with myelodysplasia-related changes (AML-MRC). These mutations placed him in the **ELN 2022 Adverse-risk** category.
- **Cytogenetics:** Normal male karyotype.

### **Initial Treatment Rationale and Course:**

At the time of diagnosis in August 2021, Mr. Watson had a borderline performance status (ECOG 2) due to profound fatigue. The VIALE-A trial data supporting Venetoclax + Azacitidine was well-established; however, the patient expressed significant concerns about the intensity, logistical burden, and infectious risks of the combination regimen. After a lengthy shared-decision making process, he opted for single-agent therapy with a hypomethylating agent, with the plan to add Venetoclax if his response was suboptimal. He began treatment with **Azacitidine** on August 13, 2021.

He experienced a slow, modest response. After 4 cycles, his transfusion needs decreased, and a repeat marrow biopsy showed a reduction in blasts to 12% (a partial remission). He maintained this stable, low-level disease for approximately one year, a period during which his quality of life was acceptable.

### **Disease Progression and Terminal Course:**

In September 2022 his platelet counts began to fall precipitously, and circulating blasts reappeared. He was admitted for a repeat bone marrow biopsy which showed aggressive disease progression with 70% blasts. His disease was now clearly refractory to Azacitidine.

We discussed adding Venetoclax as a salvage strategy. However, his clinical status had declined significantly. He was now ECOG 3-4, largely bed-bound, and had developed several comorbidities during his treatment, including a fungal skin infection and atrial fibrillation. It was felt that the risks of salvage therapy far outweighed the minimal chance of benefit.

The patient and his family agreed to transition to hospice care. He was discharged to an inpatient hospice facility and passed away peacefully from complications of progressive leukemia in February 2023, approximately 18 months after his diagnosis.



## **DISCUSSION POINTS FOR THE CONFERENCE:**

1. **Upfront Treatment Selection:** This case raises the question of treatment selection in the borderline-fit older adult with high-risk molecular features. In retrospect, given the known poor efficacy of single-agent HMA in patients with mutations like BCOR/U2AF1, should we have more strongly advocated for the upfront combination with Venetoclax despite the patient's initial reluctance?
2. **Clonal Evolution:** The progression from a stable partial remission to aggressive relapse suggests the acquisition of new mutations or clonal evolution under the selective pressure of Azacitidine. This highlights the inherent genomic instability of this disease subtype.
3. **The "Stable Disease" Dilemma:** Mr. Watson's year of stable disease on a low-intensity therapy presents a management challenge. While not a deep remission, it provided him with a year of reasonable quality life. This case forces us to balance the goals of achieving deep molecular remission versus maximizing quality time with less aggressive, but potentially less durable, treatment strategies.

**CONCLUSION:** This case is a paradigm of high-risk, secondary-type AML. It illustrates a modest but ultimately transient benefit from single-agent HMA therapy and underscores the aggressive biology conferred by high-risk molecular mutations. It serves as a powerful reminder of the importance of upfront combination therapy (Venetoclax+HMA) as the standard of care for this patient population.